SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct (192425119)

I N.R.Yasaswini ____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1.) Cloud computing is the delivery of computing services such as servers, storage, databases, analytics over the Internet on a pay-as-you-go basis. It allows users to access resources anytime without owning or maintaining physical infrastructure.

Characteristics :

1) On-Demand Self-Service : Users can access computing resources whenever required without human interaction from the service provider.

2.) Broad Network Access : Cloud services are available over the Internet and can be accessed using devices such as laptops, smartphones & tablets.

3.) Resource Pooling : Resources can be increased or decreased automatically based on user demand, making the cloud highly scalable.

4.) Rapid Elasticity : Resources are shared among multiple users using virtualization, ensuring efficient utilization and cost savings.

Difference from Traditional Computing:

Traditional systems require organization to purchase & maintain hardware, whereas cloud computing provides flexible, scalable & cost-effective resources over the Internet.

Q2) Cloud computing evolved through several technological advancements:

1) **Hardware Evolution:** Early computers were large mainframes used by multiple users through terminals.

2) **Personal Computing Era:** Personal computers became affordable, allowing individuals & businesses to own computing systems.

3) **Networking Era:** LAN & WAN technologies enabled computers to communicate & share resources.

4) **Internet Evolution:** The Internet connected millions of computers worldwide, making online communication & data sharing possible.

5) Internet Software Evolution: Technologies such as web applications, virtualization, distributed computing enabled software to run over the Internet, leading to modern cloud platforms like AWS, Microsoft Azure and Google Cloud.

Conclusion: The combination of improved hardware, networking, virtualization, and Internet technologies laid the foundation for today's cloud computing.

3.) Server virtualization is process of creating multiple virtual servers on single physical server using a hypervisor. Each virtual machine operates independently with its own OS and applications.

Role in Cloud Computing:

→ Improve hardware utilization.

→ Reduces infrastructure cost.

→ Enables rapid deployment of virtual machines.

→ Provides scalability & flexibility.

Difference :

Virtualization	Cloud Computing
→ Creates virtual versions of hardware.	→ Delivers computing services through Internet.
→ Technology used inside data centers.	→ Service model provided to customers.
→ Requires virtualization software.	→ Uses virtualization along with networking & automation.
→ Focuses on efficient resource utilization.	→ Focuses on providing scalable services on demand.

Example : VMware is used for virtualization, whereas AWS provides cloud services.

4) A data center is a facility that contains servers, storage devices, networking equipment, cooling systems and power backup used to host cloud services.

Roles of Data Centers :

→ Store & manage huge amounts of data.

→ Run applications and websites continuously.
→ Provide high availability with backup systems.
→ Ensure data security using firewalls & encryption.
→ Support disaster recovery & business continuity.
→ Enable virtualization for efficient resource sharing.

Example : Google Cloud and Microsoft Azure operate multiple data centers worldwide to provide reliable cloud serices.

5.)

Service Model	Description	Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage & networking. Users manage OS & applications.	Amazon EC2.
PaaS (Platform as a Service)	Provides a platform to develop & deploy applications without managing hardware	Google App Engine.

Saas (Software as a Service)	Delivers ready-to-use software through a web browser.	Gmail (or) Microsoft 365
Xaas (Anything as a Service).	Refers to any IT services delivered over the Internet, such as security, database, or storage services.	Dropbox, Zoom, (or) AWS

Conclusion : Iaas provides infrastructure, Paas provides a development platform, Saas provides complete software applications, and XaaS is a ~~ba~~ board category covering all cloud-based Services.